NASA TechRise Post-Flight Analysis

EXPERIMENT TITLE

UV Absorption of Different Gasses and Compositions.

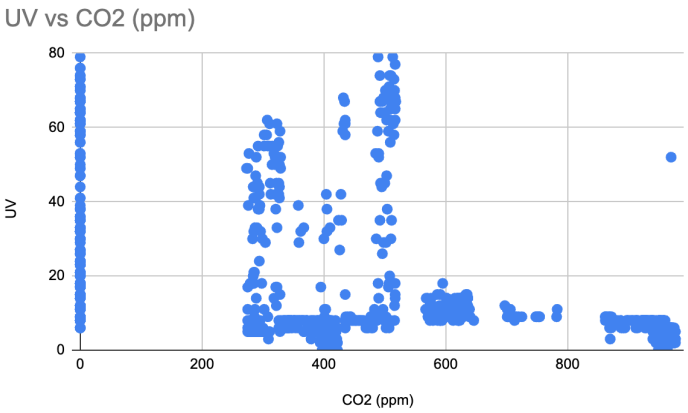
School Name: Montgomery High School **Team #** (e.g., World 1): World 33

Findings: Summarize the goal/hypothesis of the experiment. What did you learn from your experiment? Was your hypothesis correct? Why or why not? Did you learn anything unexpected? You may reference graphs and images you add to the section below as necessary.
***Reminder: Use clear and concise language to explain what your team observed.*

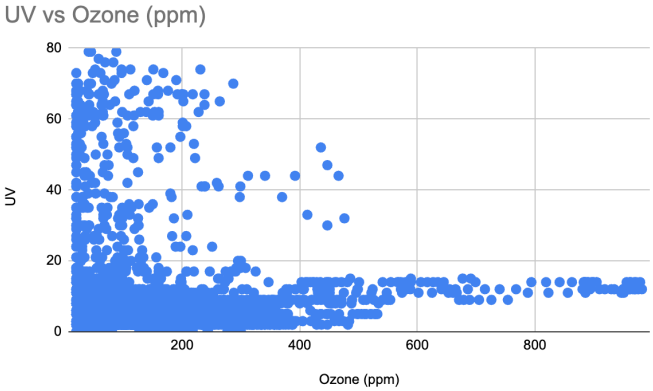
The goal of our experiment was to find any possible correlation between different gasses and their absorption of UV radiation. Our prediction was that the gasses with the highest densities will absorb more UV radiation than less dense gasses, in addition to the fact that some gasses may have more of an effect on it than others due to their nature. Our experiment yielded the expected result that more ozone particles results in greater UV absorption. At the same time we realized during our analysis that if there was a relationship between any of the components of the atmosphere and UV radiation, our graphs would reflect it as if the UV was getting affected by the measured gas whereas in reality it would be effected by something else entirely. We found a general trend between gasses and UV as the ppm of the gasses increased, the level of UVS suddenly decreased in the graph. When the ppm of the gas was near zero, the UVS where concentrated toward the higher spectrum of the graph. Overall, we were hoping to see a clear cut relationship between the gasses and the UV absorption but in fact a major flaw prevented us from observing one.

Graphs and Images: Insert a graph illustrating the trends from your data or any exemplary photos from your experiment.

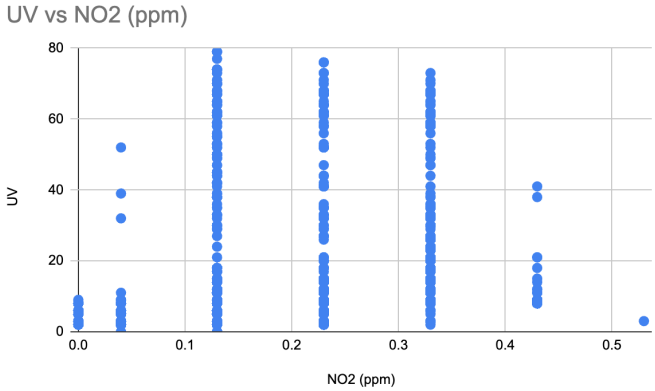
***Note: Any additional photos can be placed in your 'Team Raw Data' folder.*



Trend: As the CO2 composition density continues, the UV radiations follows a decreasing trend. This would suggest that CO2 concentrations does effect UV radiation



Trend: As the Ozone concentration increases, the UV radiation follows a decreasing trend. This supports the idea that higher ozone concentrations do reduce UV radiation.



Trend: As the NO2 concentration increases, the UV radiation doesn't change on the basis of a pattern. This suggests that the concentration of NO2 doesn't effect the UV radiation.

Reflection: If you had the chance to redo your experiment, what would you do differently? What are you now curious to learn, explore, or study based on your data findings?

If we had to redo the experiment, we would change certain elements of our process in order to allow for a data collection that can be analyzed. After reflection, we found the original hypothesis very shaky and not precise. What we mean by that is that we should have accounted for problems that could potentially come up during data analysis based on our data collection. Generally we should have done more research beforehand. Since we didn't, it led to a disorganized structure of the experiment. We were not able to detect the percentage composition of the gasses, but rather the individual concentrations of the gasses. Instead, we should have adapted our hypothesis throughout the experiment according to our available materials. This was effectively a testing experiment to see whether UV radiation was truly absorbed by ozone particles.

We are definitely excited to create a new experiment to submit to the techrise challenge again. Our team is even more eager to create another project due to the experience we have gathered from our current experiment. We plan to come up with a more meaningful and thorough experiment. Additionally, we will also be able to work on it more efficiently owing to the fact that we have gained a basic understanding of certain skills such as soldering that were necessary but took a long time to learn. This way we can focus more on the end goal rather than the skills needed to get there, although we are well aware that different skills are required for different projects. Overall we feel more inclined to work on another project as a result of our experience from this current one which has piqued our interest in this type of student-led project.

Disclaimer: This experiment was a part of NASA TechRise 2023-24 Challenge, funded and flight tested by the NASA's Flight Opportunities Program. However, NASA has not independently verified or validated any data, results, or conclusions arising from this research. NASA expressly disclaims any liability or responsibility for the accuracy of information or drawn conclusions herein.